Hybrid NLP & Behavioral Classification Engine

Technical Report

Submitted by
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1 Introduction

The Hybrid NLP & Behavioural Classification Engine is a multiclass machine learning system designed to identify artworks from student survey responses. As part of the project, data collection was conducted by gathering responses from students who were shown a series of artworks and asked to record their reactions. The project combines natural language processing (NLP) techniques with structured behavioural data to investigate whether subjective emotional responses can be used to distinguish between different visual artworks.

The dataset contains both structured survey responses, such as emotion intensity ratings and perceived colour counts, and unstructured free-response text describing feelings, food analogies, environmental associations, and imagined soundtracks. Because the data was collected directly from students, it reflects naturally occurring variation in interpretation, language use, and emotional perception, creating a challenging classification problem involving heterogeneous data types.

The primary objective of the project is to transform noisy human responses into meaningful machine learning features and evaluate multiple classification approaches. Particular emphasis is placed on text feature engineering, model comparison, hyperparameter tuning, and implementing multiclass logistic regression from scratch using NumPy.

2 Project Pipeline

The overall project workflow follows a standard machine learning pipeline: raw survey responses are cleaned, transformed into numerical features, used to train several candidate models, and evaluated on held-out validation and test data.

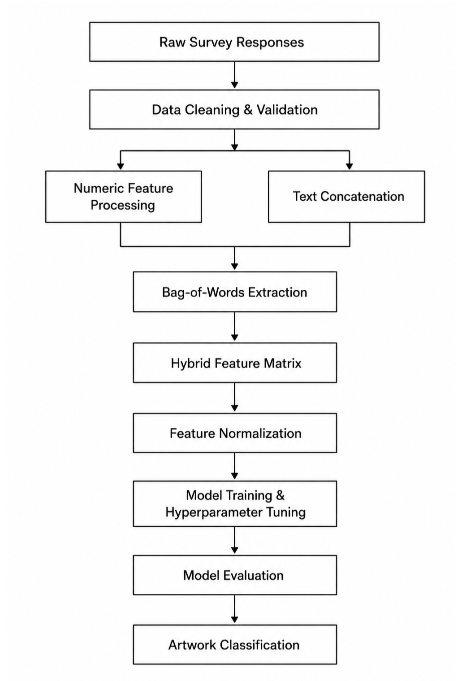

Figure 1: End-to-end machine learning pipeline.

3 Dataset and Data Preparation

The dataset consists of survey responses associated with three artwork classes. Each observation contains a target painting label, structured numeric responses, and free-response text fields. The numeric features include emotion intensity, agreement ratings for emotional descriptors, number of perceived colours, number of perceived objects, and estimated willingness to pay. The text features include descriptions of feelings, room placement preferences, social viewing preferences, seasonal associations, food analogies, and imagined soundtracks.

Because the responses were collected from open-ended human input, the raw dataset contained missing values, inconsistent formatting, and outliers. Several numeric fields required cleaning before they could be used for model training. Rating-based columns were converted into integer values, while text-based numeric responses, such as willingness to pay, were parsed using numeric extraction.

Missing numeric values were replaced using median imputation. Outliers in features such as prominent colours, objects noticed, and willingness to pay were handled using an interquartile range approach. Values outside the acceptable range were replaced with the median rather than removed, preserving the number of training examples while reducing the influence of unrealistic responses.

The dataset was then shuffled and split into training, validation, and test sets using a 70% / 15% / 15% split. The training set was used to learn model parameters, the validation set was used for model selection and hyperparameter tuning, and the test set was reserved for final evaluation.

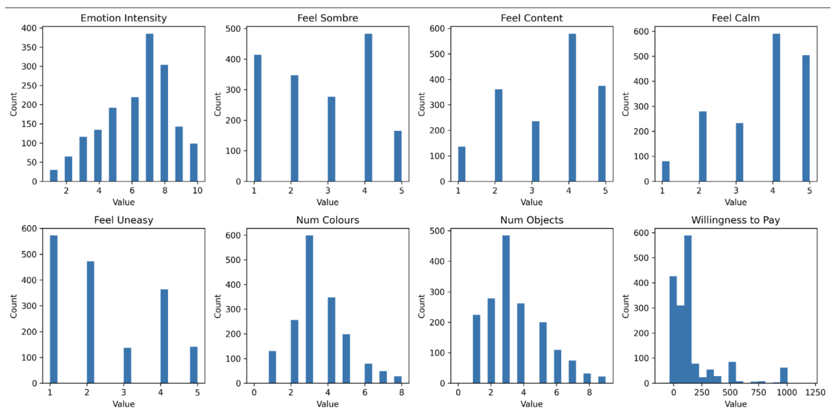

Figure 2: Distribution of cleaned numeric survey features.

4 Feature Engineering

The dataset contains both structured survey responses and unstructured free-text responses. While the numeric features provide information regarding emotional intensity, colour perception, object perception, and willingness to pay, exploratory analysis suggested that the text responses contained the strongest predictive signals.

To utilize the text data effectively, all text-based survey fields were concatenated into a single document for each respondent. These documents were then preprocessed through tokenization and converted into a Bag-of-Words representation. Under this representation, each unique word in the vocabulary corresponds to a feature, and each response is represented as a sparse vector indicating word presence.

This approach allowed the model to learn associations between descriptive language and specific artwork classes. For example, certain words related to dreams, night, water, calmness, or uneasiness may be more strongly associated with particular artworks. Although Bag-of-Words does not preserve word order or deeper semantic meaning, it is simple, interpretable, and effective for many classical text classification tasks.

The Bag-of-Words transformation produced a high-dimensional sparse feature space, where each unique word in the vocabulary corresponds to a separate feature. As a result, most entries in the feature matrix were zero because individual responses contained only a small subset of the overall vocabulary. This sparsity is common in text classification problems and was one of the primary reasons logistic regression performed well, as linear models are often effective in high-dimensional sparse feature spaces.

The resulting text features were combined with the cleaned numeric survey features to create a hybrid feature representation. Both text and numeric features were normalized using statistics computed from the training set. This ensured that no feature group dominated the model purely due to scale differences.

Experimental results showed that text features contributed substantially more predictive power than numeric features alone. Models trained primarily on Bag-of-Words features consistently achieved higher validation accuracy, indicating that descriptive language and emotional associations were highly informative for distinguishing between artwork classes.

5 Model Exploration

Several classification models were evaluated to determine which approach performed best on the survey dataset. The main models tested were logistic regression, random forest, random forest combined with Naive Bayes probabilities, and decision tree classification.

Initial experiments using primarily numeric features produced weaker results, which aligned with the exploratory analysis. Many numeric survey features were clustered around moderate values and overlapped heavily across artwork classes. This made it difficult for models relying only on structured numeric data to distinguish between paintings.

To improve predictive performance, text features were incorporated using the Bag-of-Words representation. This created a high-dimensional and sparse feature space. Logistic regression performed well in this setting because linear models are often effective for sparse text classification problems. In contrast, tree-based models such as decision trees and random forests struggled more with the sparse representation and showed less consistent validation performance.

A hybrid random forest and Naive Bayes approach was also tested. In this setup, Naive Bayes was trained on the text features and its output probabilities were used as additional inputs to the random forest. Although this improved over using numeric features alone, it did not outperform logistic regression.

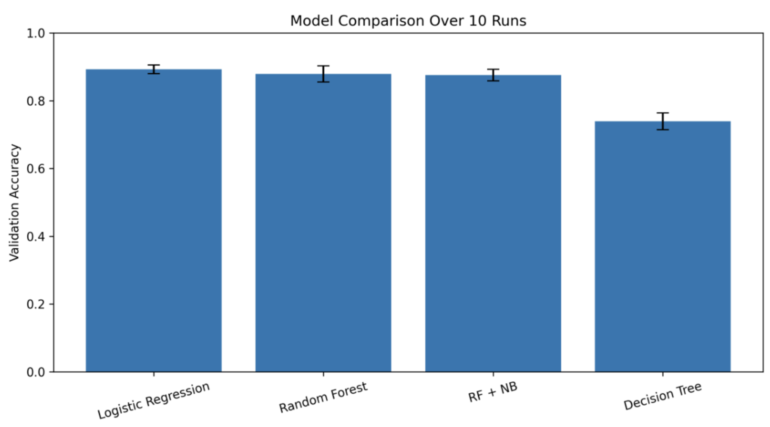

Figure 3: Validation accuracy comparison across candidate models.

6 Final Model and Training

The final model selected was multiclass logistic regression trained on the hybrid feature representation. Logistic regression was chosen because it performed consistently well on the validation set, trained efficiently, and was well suited to the sparse high-dimensional Bag-of-Words feature space.

The model was implemented from scratch using NumPy rather than relying solely on a library implementation. The implementation includes softmax-based multiclass classification, cross-entropy loss optimization, vectorized gradient descent, L2 regularization, and prediction using the class with maximum probability.

For each sample, the model computes class scores using

$$Z = XW^T + b$$

where X is the feature matrix, W is the learned weight matrix, and b is the bias vector. These scores are converted into class probabilities using the softmax function:

$$P(y = k \mid x) = \frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}$$

The model is trained by minimizing cross-entropy loss. During each training iteration, predictions are computed, the loss is evaluated, gradients are calculated, and the weights are updated using gradient descent. L2 regularization is added to reduce overfitting by discouraging excessively large weights.

Building the model from first principles provided direct control over the training process and improved the interpretability of the optimization procedure. It also allowed the implementation to be connected directly to the feature engineering pipeline and saved model artifacts for inference.

7 Hyperparameter Tuning

Several hyperparameters were tuned using validation performance, including learning rate, number of training iterations, vocabulary frequency threshold, and L2 regularization strength.

The learning rate controls the size of each gradient descent update. A very small learning rate can converge slowly, while a large learning rate can produce unstable training. Several learning rates were evaluated using validation accuracy as the primary selection criterion. While the highest validation accuracy was observed at a learning rate of 0.01, larger learning rates produced comparable performance while converging more quickly. A learning rate of 0.05 was ultimately selected as a practical balance between optimization efficiency and predictive accuracy.

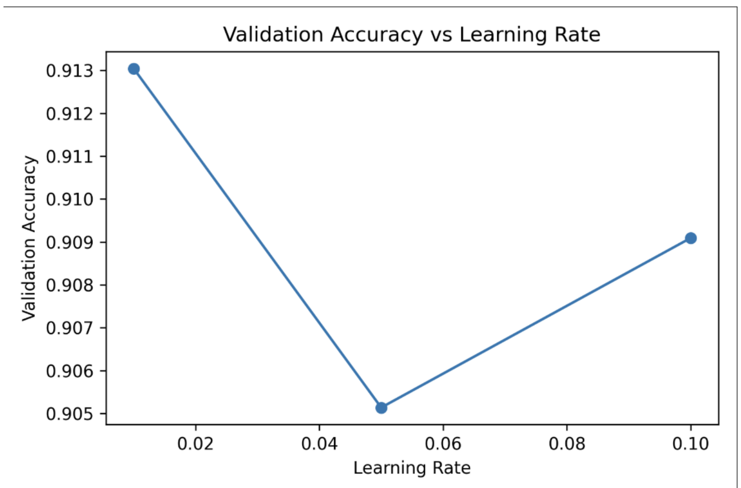

Figure 4: Validation accuracy across learning rate values.

L2 regularization was also evaluated to improve model generalization and reduce overfitting. Validation results indicated that introducing regularization improved performance relative to no regularization. Performance remained relatively stable across several positive regularization strengths, suggesting that the model was not highly sensitive to the exact choice of λ within the tested range. A value of $\lambda = 0.5$ was selected for the final model.

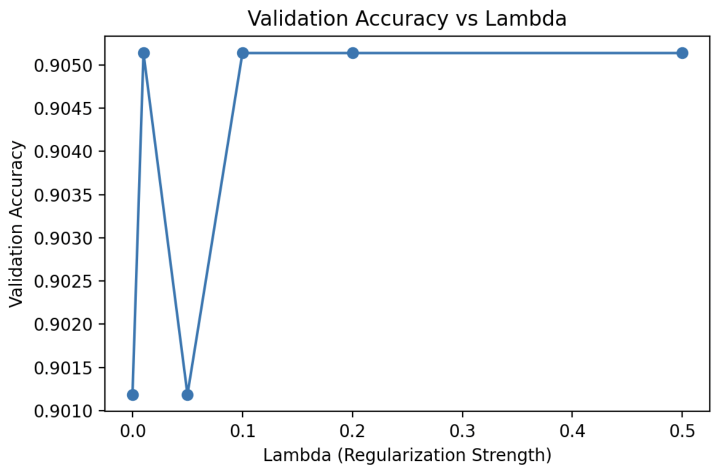

Figure 5: Validation accuracy across L2 regularization strengths.

The final selected hyperparameters were:

- Learning rate: (0.05)
- Number of iterations: (3000)
- L2 regularization strength: ($\lambda = 0.5$)
- Feature representation: Bag-of-Words text features combined with cleaned numeric survey features

8 Results and Discussion

The final logistic regression model achieved approximately 89% training accuracy and 91% validation accuracy. This result suggests that the model was able to learn meaningful relationships between survey responses and artwork classes without severe overfitting.

The model comparison results showed that logistic regression outperformed the tested tree-based approaches. This outcome is consistent with the structure of the data: the strongest predictive signals came from text responses, and Bag-of-Words representations produce sparse high-dimensional feature spaces where linear classifiers often perform well.

The results also showed that numeric features alone were insufficient for achieving strong classification performance. Emotional ratings, colour counts, object counts, and willingness-to-pay estimates provided some signal, but they were less discriminative than free-response text. The strongest performance came from combining natural language features with structured behavioural responses.

Overall, the project demonstrates the importance of feature engineering when working with heterogeneous survey data. Proper text processing, numeric cleaning, normalization, and model selection were all necessary to produce a reliable classification pipeline.

9 Conclusion

Hybrid NLP & Behavioural Classification Engine demonstrates how noisy human survey responses can be transformed into useful machine learning features for multiclass classification. By combining cleaned numeric features with Bag-of-Words text representations, the project captures both structured behavioural signals and unstructured emotional descriptions.

The final system uses a custom NumPy implementation of multiclass logistic regression with softmax, cross-entropy loss, gradient descent, and L2 regularization. Model comparison experiments showed that logistic regression was best suited for the sparse text-driven feature space, outperforming decision tree and random forest alternatives.

The project highlights the value of data cleaning, NLP feature engineering, and careful model evaluation in building practical machine learning systems from real-world human response data.